Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.338701332008072, median 0.27370237689458043, std: 0.25410458424268595
Gyroscope error (imu0): mean 0.2499015827295168, median 0.19151679459379453, std: 0.26949739934869155
Accelerometer error (imu0): mean 0.24689642783838903, median 0.18818760460637743, std: 0.26443172825226186

Residuals

Reprojection error (cam0) [px]: mean 0.338701332008072, median 0.27370237689458043, std: 0.25410458424268595
Gyroscope error (imu0) [rad/s]: mean 0.001941365350495385, median 0.0014878019778880715, std: 0.002093595836527574
Accelerometer error (imu0) [m/s^2]: mean 0.038699429714072635, median 0.029497198648379252, std: 0.04144797545782076

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99896808 -0.0453126 -0.00309015 0.00064399]
 [0.00390519 0.15348194 -0.98814374 -0.00066731]
 [0.04524964 0.98711198 0.15350051 -0.03217009]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.99896808 0.00390519 0.04524964 0.00081496]
 [-0.0453126   0.15348194 0.98711198 0.03188708]
 [-0.00309015 -0.98814374 0.15350051 0.00428072]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.009685103117698813

Gravity vector in target coords: [m/s^2]
[0.005498   -9.8045301 -0.19895266]

Calibration configuration
==========

cam0

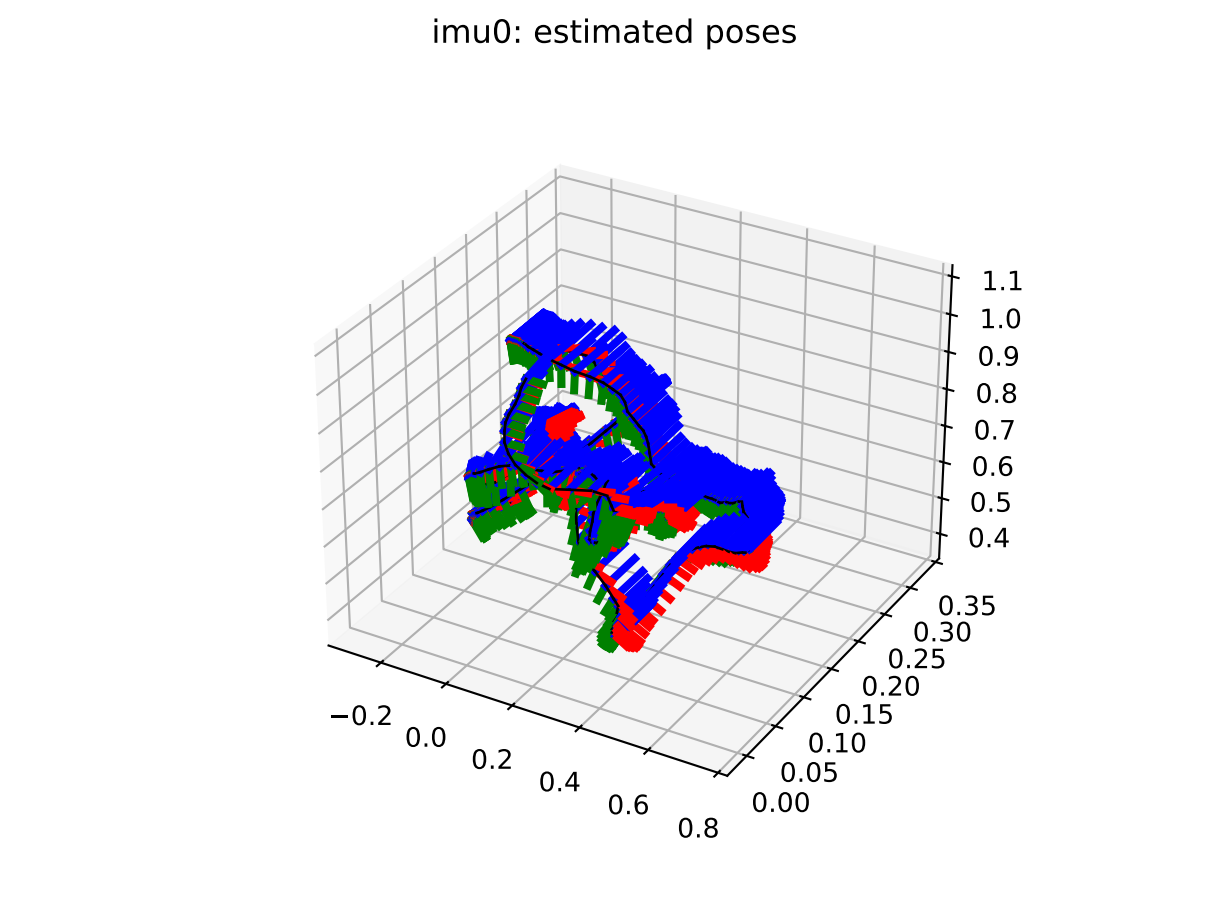
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

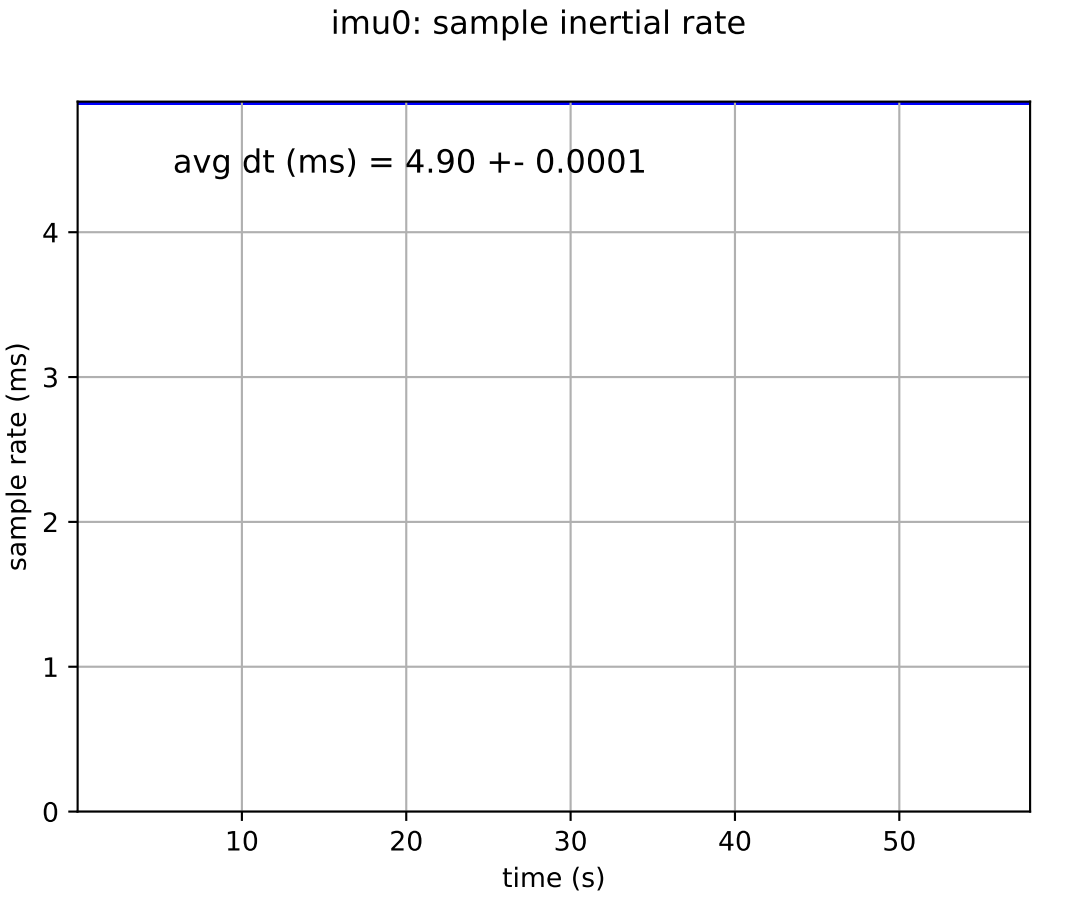
IMU configuration
==========

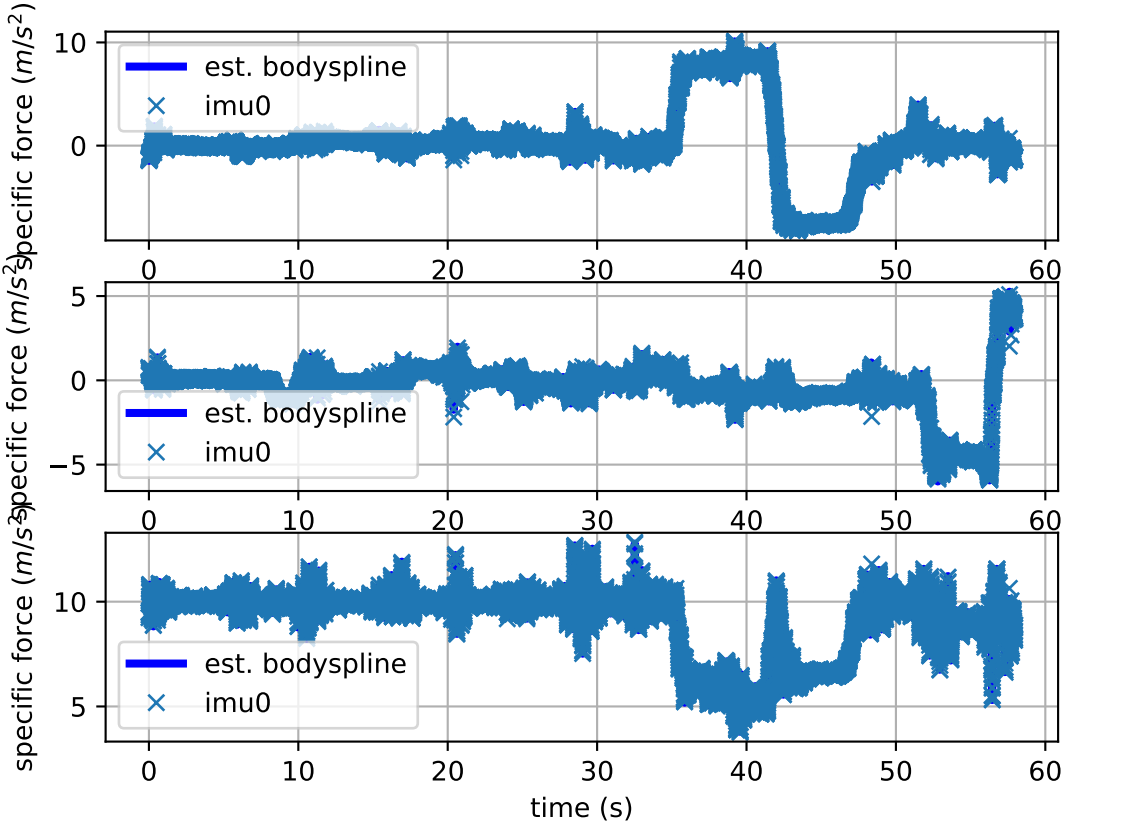
IMU0:

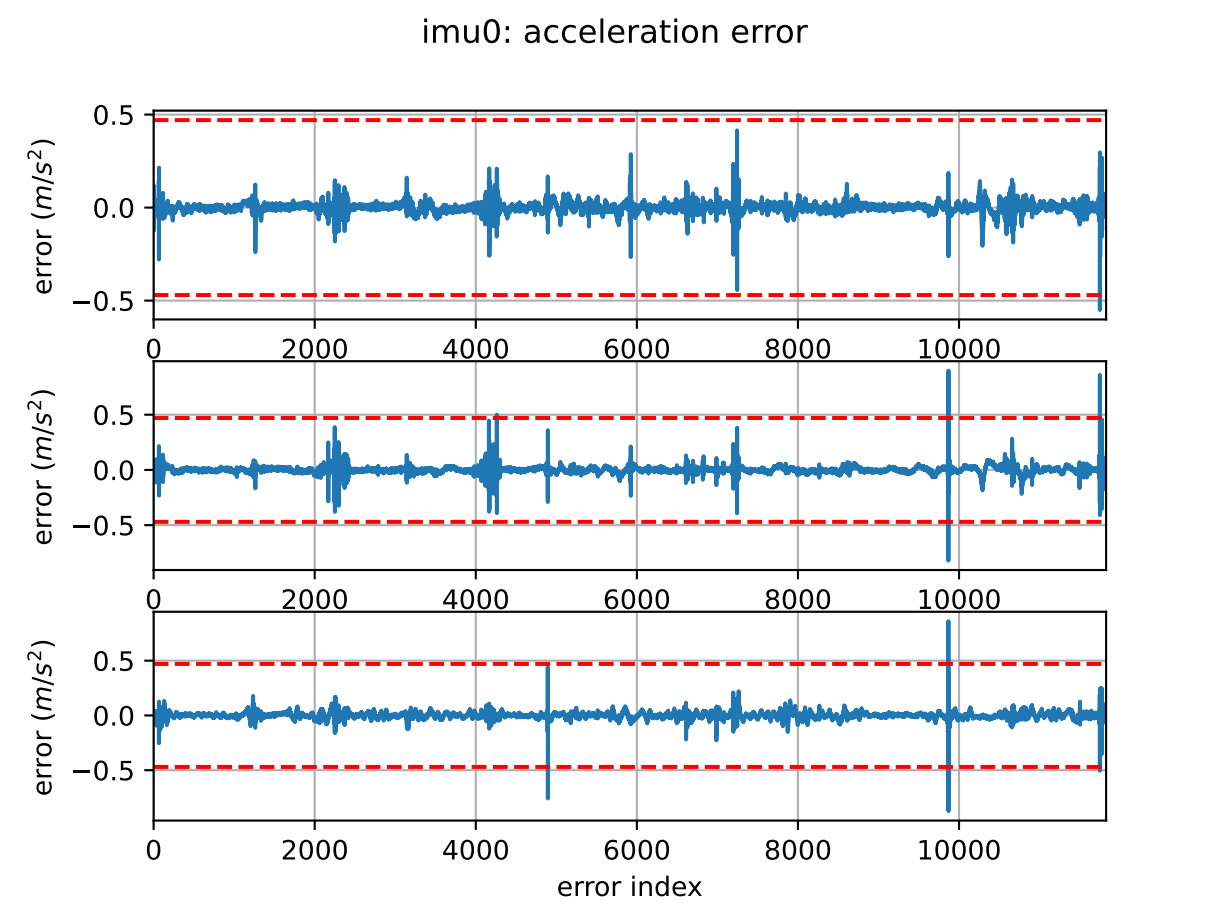
  Model: calibrated
  Update rate: 204.0355927763645
  Accelerometer:
    Noise density: 0.010973288353646839
    Noise density (discrete): 0.15674357888808385
    Random walk: 0.006539909582683411
  Gyroscope:
    Noise density: 0.0005438577233174098
    Noise density (discrete): 0.00776851962797146
    Random walk: 5.5863724525824024e-05

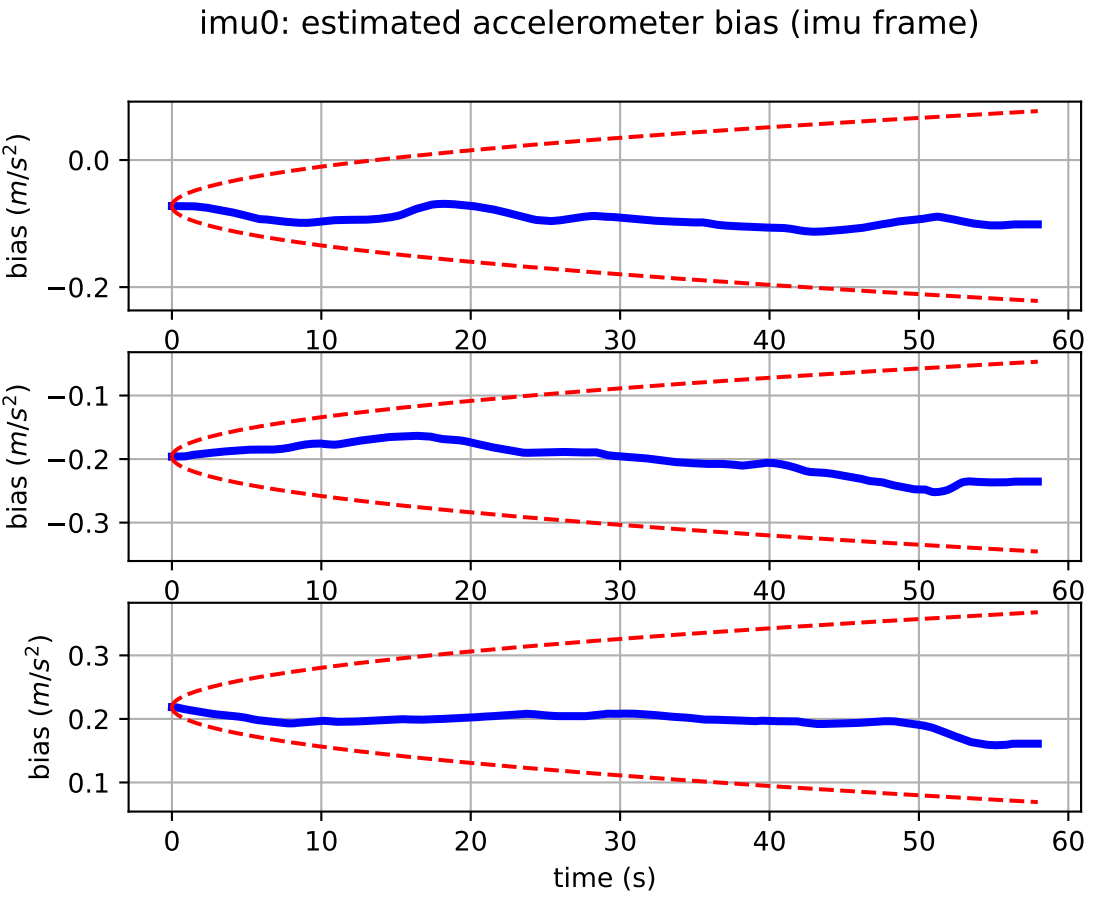
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

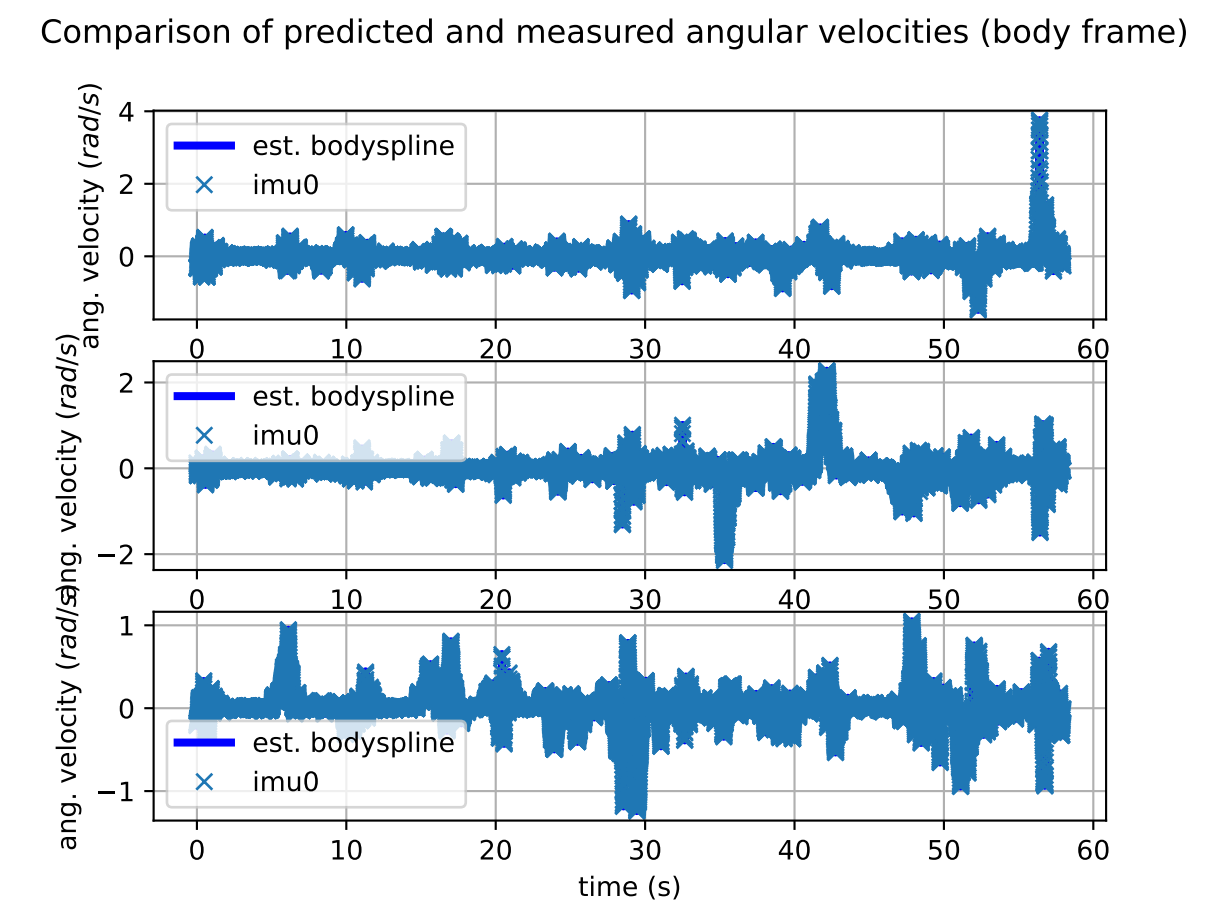

imu0: estimated poses
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.2
0.0
0.2
0.4
0.6
0.8

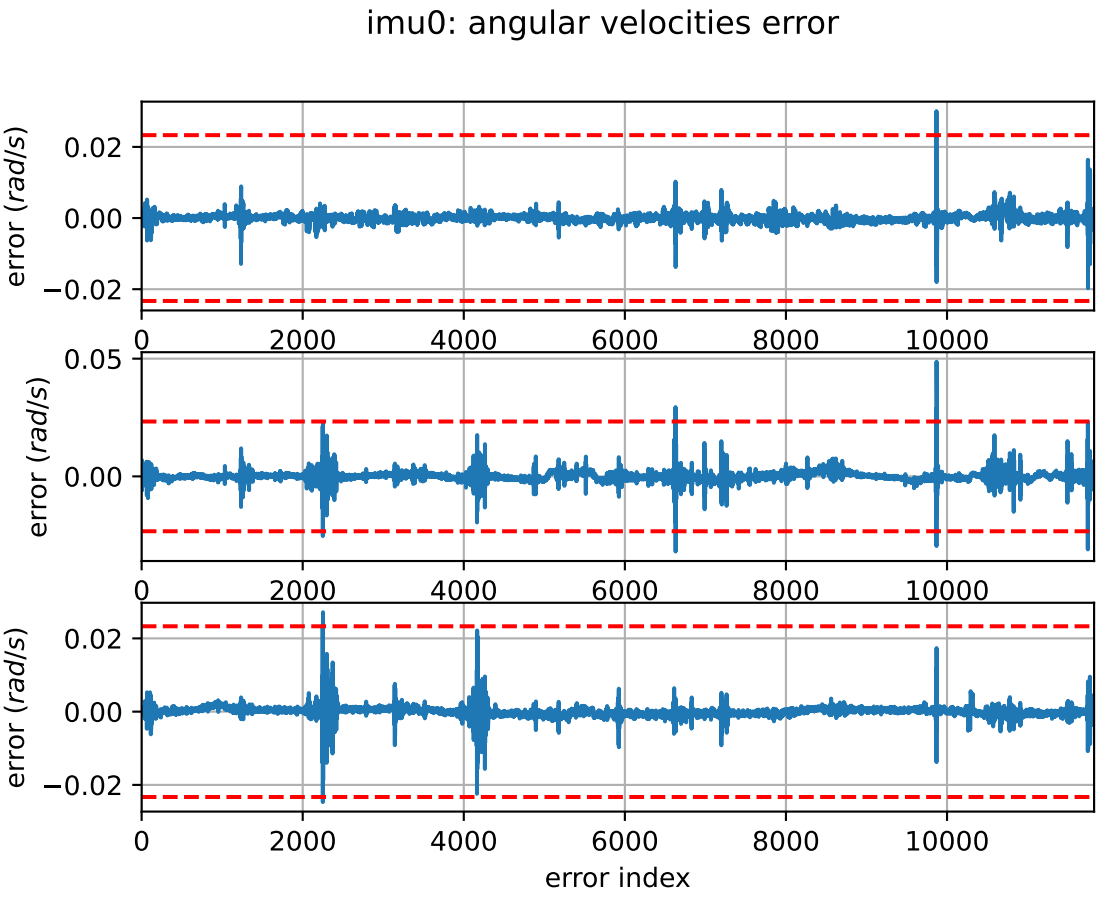

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0001
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

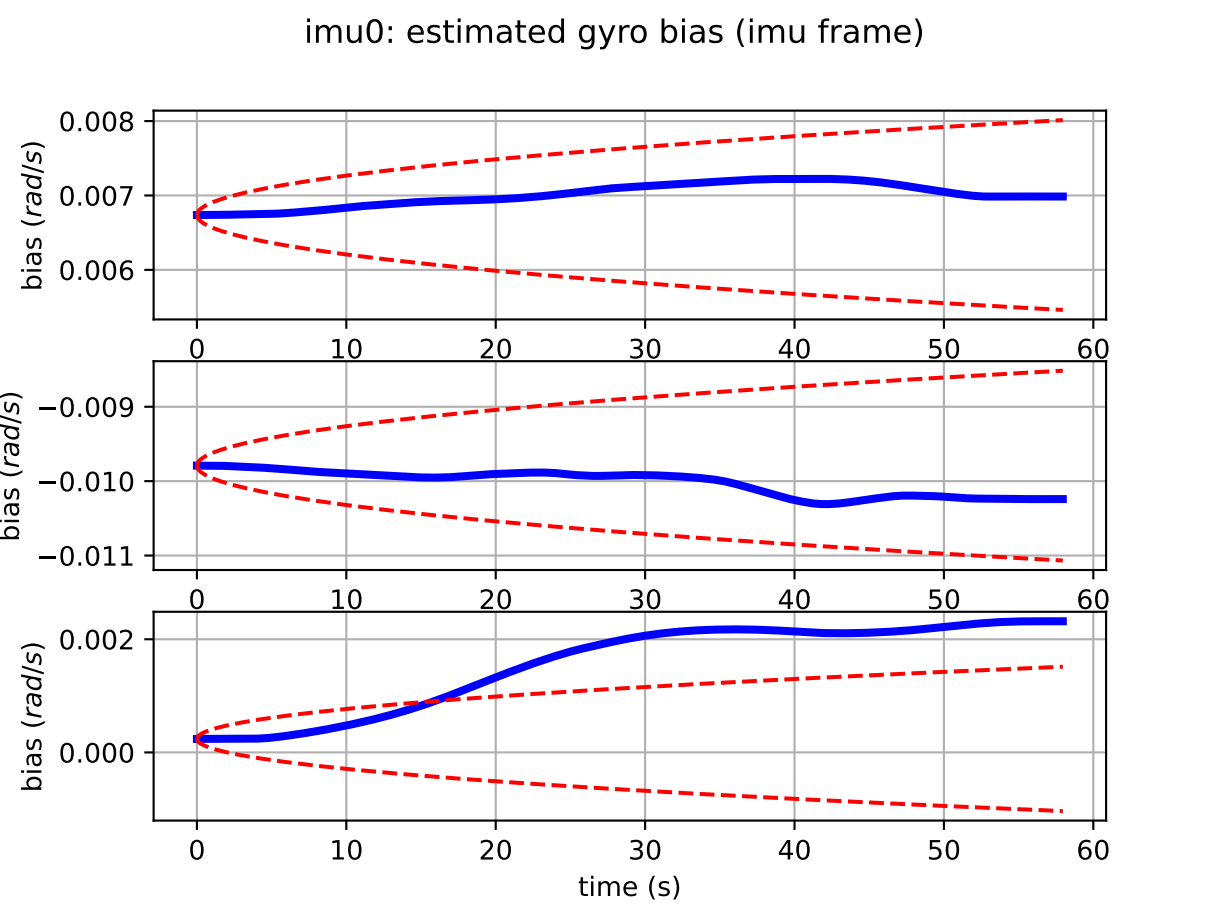

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
time (s)

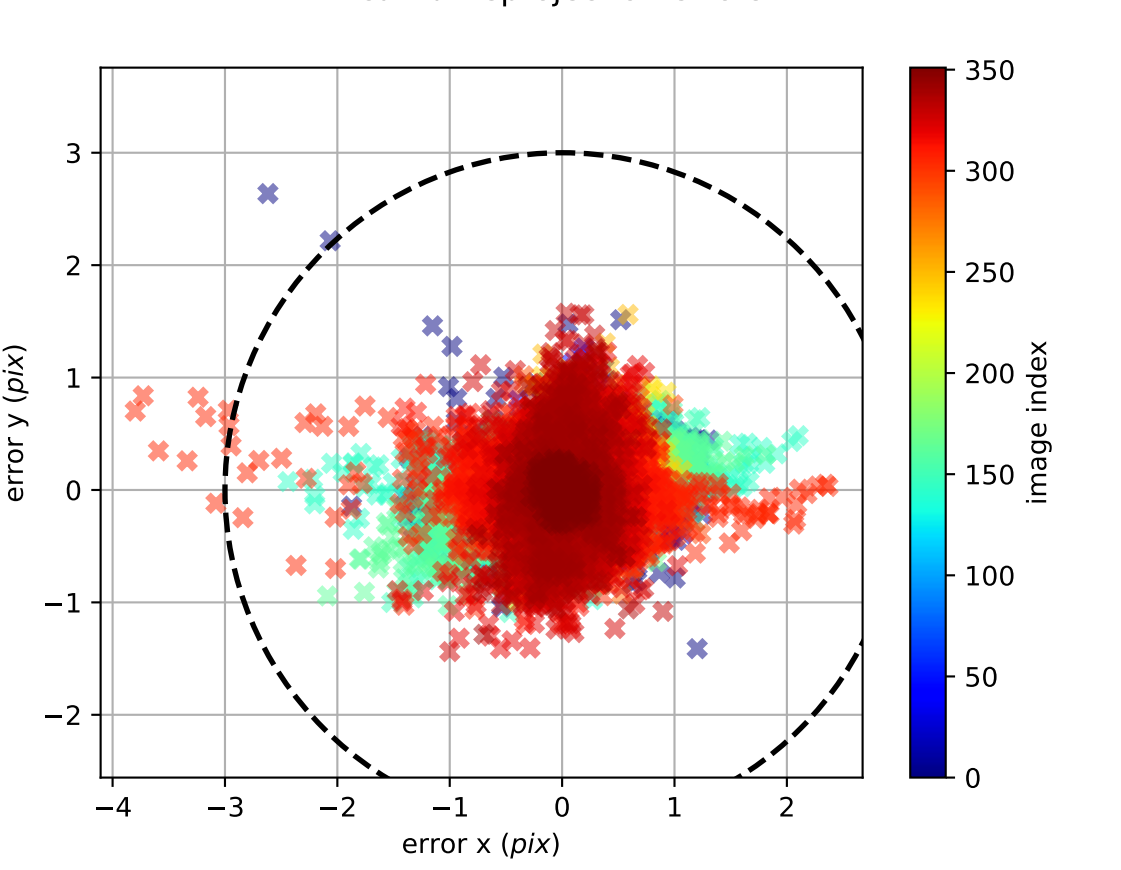

imu0: acceleration error
error (m/s^2)
0.5
0.0
−0.5
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.0
−0.2
0
10
20
30
40
50
60
bias (m/s^2)
−0.1
−0.2
−0.3
0
10
20
30
40
50
60
bias (m/s^2)
0.3
0.2
0.1
0
10
20
30
40
50
60
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
0.02
0.00
−0.02
0
2000
4000
6000
8000
10000
error (rad/s)
0.05
0.00
error (rad/s)
0.02
0.00
−0.02
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.008
0.007
0.006
0
10
20
30
40
50
60
bias (rad/s)
−0.009
−0.010
−0.011
bias (rad/s)
0.002
0.000
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−4
−3
−2
−1
0
1
2
3
0
50
100
150
200
250
300
350